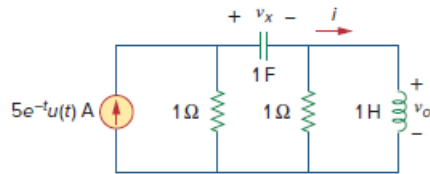


Problem

Find $v_o(t)$ in the circuit of Fig. 16.82 if $v_x(0) = 10$ V and $i(0) = 5$ A.

Figure 16.82:



Step-by-step solution

Step 1 of 3

Refer to Figure 16.82 in the text book.

The Laplace transform of source voltage $5e^{-t}u(t)$ is $\frac{5}{s+1}$.

Determine the impedance of capacitor 1 F in Laplace transform.

$$\begin{aligned}\frac{1}{sC} &= \frac{1}{s(1)} \\ &= \frac{1}{s}\end{aligned}$$

Determine the impedance of capacitor 1 H in Laplace transform.

$$\begin{aligned}sL &= s(1) \\ &= s\end{aligned}$$

Step 2 of 3

Apply Kirchhoff's current law at node 1.

$$\begin{aligned}\frac{5}{s+1} &= \frac{v_x}{1} + \frac{v_x}{(1/s)} \\ &= sv_x + v_x \\ &= v_x(s+1) \\ v_x(s) &= \frac{5}{(s+1)^2} \dots\dots\dots (1)\end{aligned}$$

Apply Kirchhoff's current law at node 2.

$$\begin{aligned}\frac{v_o}{s} + \frac{v_o}{1} &= \frac{v_x}{(1/s)} \\ sv_o &= v_o(1 + 1/s) \\ v_x(s) &= v_o\left(\frac{1}{s} + \frac{1}{s^2}\right) \dots\dots\dots (2)\end{aligned}$$

Step 3 of 3

Substitute equation (1) in (2).

$$\frac{5}{(s+1)^2} = v_0 \left(\frac{1}{s} + \frac{1}{s^2} \right)$$

$$v_0(s) = \frac{5s^2}{(s+1)^3}$$

Express the $v_o(s)$ in partial fractions.

$$\begin{aligned} v_0(s) &= \frac{5s^2}{(s+1)^3} \\ &= \frac{5}{s+1} - \frac{10}{(s+1)^2} \end{aligned}$$

Applying inverse Laplace transform we get

$$v_0(t) = \left[5e^{-t} - 10e^{-\frac{t}{2}} \cos(t/2) \right] u(t) \text{ V}$$

Therefore the voltage across the inductor is $\boxed{5e^{-t} - 10e^{-\frac{t}{2}} \cos(t/2)} u(t) \text{ V}.$

[Comments \(1\)](#)



Anonymous

Missing a third term on partial fraction
